7 **(a)** **(i)** mass defect = [92 × 1.007 276 + (235 – 92) × 1.008665 – 235.043930] u M1
= 1.864 557 u

binding energy = 1.864 557 × 1.66 × 10^{-27} × (3.00 × 10^{8})^{2} / 1.60 × 10^{-19} M1
= 1.741,030 · 10^{9} eV = 1741.030 MeV

binding energy per nucleon = 1741.030 / 235 M1

= 7.409 MeV A0

(ii) less (than) B1

(b) energy = (95 × 8.443) + (139 × 8.189) – (235 × 7.409) M1

= 199 MeV or 199.5 MeV A1

(c) number of reaction = 1.2 × 10^{-7} × 6.02 × 10^{23} C1
= 7.224 × 10^{16}

power = 7.224 × 10^{16} × 199.5 × 10^{6} × 1.60 × 10^{-19} / 25 × 10^{-3} M1

= 9.2 to 9.22 × 10^{7} W A1